

CURSOR

```
mysql> use cursors;
Database changed
mysql> CREATE TABLE students (
->     id INT,
->     name VARCHAR(50),
->     mark INT
-> );
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> INSERT INTO students VALUES
-> (1, 'Arun', 85),
-> (2, 'Bala', 90),
-> (3, 'Cathy', 78);
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE StudentCursor()
-> BEGIN
->     DECLARE done INT DEFAULT 0;
->     DECLARE student_name VARCHAR(50);
->     DECLARE student_mark INT;
->
->     DECLARE cur CURSOR FOR
->         SELECT name, mark FROM students;
->
->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
->
->     OPEN cur;
->
->     read_loop: LOOP
->         FETCH cur INTO student_name, student_mark;
->
->         IF done = 1 THEN
->             LEAVE read_loop;
->         END IF;
->     END IF;
```

TRIGGERS

BEFORE INSERT

```
mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER before_student_insert
-> BEFORE INSERT ON students
-> FOR EACH ROW
-> BEGIN
->     IF NEW.mark < 0 THEN
->         SET NEW.mark = 0;
->     END IF;
-> END //
Query OK, 0 rows affected (0.08 sec)

mysql>
mysql> DELIMITER ;
mysql> INSERT INTO students VALUES (1, 'Arun', -10);
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> SELECT * FROM students;
+----+-----+-----+
| id | name | mark |
+----+-----+-----+
|  1 | Arun |    0 |
+----+-----+-----+
1 row in set (0.00 sec)
```

AFTER INSERT

```
mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER after_student_insert
-> AFTER INSERT ON students
-> FOR EACH ROW
-> BEGIN
->     INSERT INTO student_log(student_id, action, new_mark)
->     VALUES (NEW.id, 'INSERT', NEW.mark);
-> END //
Query OK, 0 rows affected (0.09 sec)

mysql>
mysql> DELIMITER ;
mysql> INSERT INTO students VALUES (2, 'Bala', 85);
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> SELECT * FROM student_log;
+-----+-----+-----+-----+-----+
| id | student_id | action | old_mark | new_mark |
+-----+-----+-----+-----+-----+
| 1 |          2 | INSERT |    NULL |        85 |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

BEFORE UPDATE

```
mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER before_student_update
-> BEFORE UPDATE ON students
-> FOR EACH ROW
-> BEGIN
->     IF NEW.mark > 100 THEN
->         SET NEW.mark = 100;
->     END IF;
-> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> UPDATE students
-> SET mark = 150
-> WHERE id = 2;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SELECT * FROM students;
+-----+-----+-----+
| id | name | mark |
+-----+-----+-----+
| 1 | Arun |    0 |
| 2 | Bala |  100 |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

AFTER UPDATE

```
mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER after_student_update
-> AFTER UPDATE ON students
-> FOR EACH ROW
-> BEGIN
->     INSERT INTO student_log(student_id, action, old_mark, new_mark)
->     VALUES (NEW.id, 'UPDATE', OLD.mark, NEW.mark);
-> END //
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> DELIMITER ;
mysql> UPDATE students
-> SET mark = 95
-> WHERE id = 2;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SELECT * FROM student_log;
+----+-----+-----+-----+-----+
| id | student_id | action | old_mark | new_mark |
+----+-----+-----+-----+-----+
| 1  |          2 | INSERT |     NULL |      85 |
| 2  |          2 | UPDATE |      100 |      95 |
+----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

BEFORE DELETE

```
mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER before_student_delete
-> BEFORE DELETE ON students
-> FOR EACH ROW
-> BEGIN
->     INSERT INTO student_log(student_id, action, old_mark)
->     VALUES (OLD.id, 'DELETE', OLD.mark);
-> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> DELETE FROM students
-> WHERE id = 2;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> SELECT * FROM student_log;
+----+-----+-----+-----+-----+
| id | student_id | action | old_mark | new_mark |
+----+-----+-----+-----+-----+
| 1  |          2 | INSERT |     NULL |      85 |
| 2  |          2 | UPDATE |      100 |      95 |
+----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```